

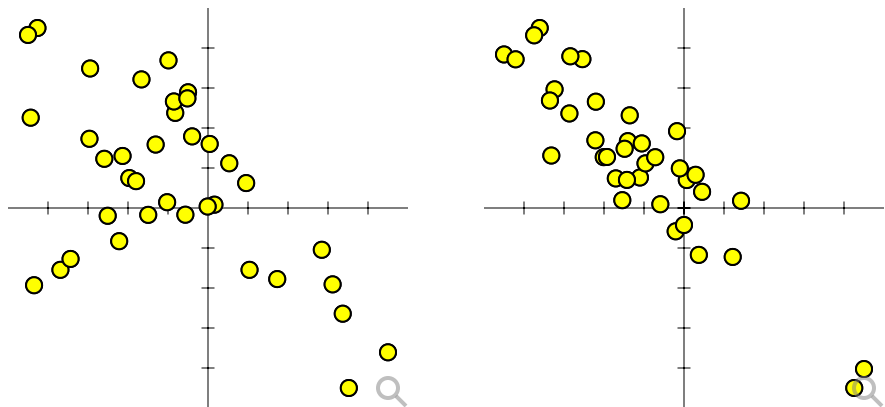
EXAM 3

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Question 1

1 pt

Here are the scatter plots for two sets of bivariate data with the same response variable. The first compares the variables  $x$  &  $y$ . The second compares the variables  $w$  &  $y$



- Which explanatory variable has a stronger relationship with the response variable  $y$ ?
- ☐ The first variable ( $x$ ) has a stronger relationship with the response variable  $y$ .
  - ☐ The second variable ( $w$ ) has a stronger relationship with the response variable  $y$ .

Question 2

1 pt

Fill out the table: (Round the answers to 2 decimal places)

| $x$         | $y$         | $x - \bar{x}$ | $(x - \bar{x})^2$ | $y - \bar{y}$ | $(y - \bar{y})^2$ | $(x - \bar{x})(y - \bar{y})$ |
|-------------|-------------|---------------|-------------------|---------------|-------------------|------------------------------|
| 4           | 1           | 1.5           | 2.25              | -0.5          | 0.25              | -0.75                        |
| 1           | 2           |               |                   |               |                   |                              |
| $\bar{x} =$ | $\bar{y} =$ | $S_{xx} =$    |                   | $S_{yy} =$    |                   | $S_{xy} =$                   |

Question 3

1 pt

Fill in the missing values for this ANOVA source table, and use it to calculate the effect size:

|         | <i>SS</i> | <i>df</i> | <i>MS (S<sup>2</sup>)</i> | <i>F</i> |
|---------|-----------|-----------|---------------------------|----------|
| Between | 473.508   | 4         |                           |          |
| Subject | 405.888   | 4         |                           |          |
| Error   |           |           |                           |          |
| TOTAL   | 1215.396  |           |                           |          |

$R^2 =$  \_\_\_\_\_

#### Question 4

✓ 1 pt ↻ 1

What is the  $F_{crit}$  value for a between groups study with 7 groups, having 6 participants per group, and a level of significance ( $\alpha$ ) of 0.1?

Use software such as Excel if your table is not specific enough:

$F.INV.RT(\alpha, df_{\text{between}}, df_{\text{within}})$ .

If using software, round to two decimal places.

$F_{crit} =$  \_\_\_\_\_

#### Question 5

✓ 1 pt ↻ 1

Which of the following are tools used in Statistical Process Control (SPC)?

- ☐ q-q plot
- ☐ Venn diagram
- ☐ k-plot
- ☐ t-test
- ☐ Check sheet
- ☐ Pareto chart
- ☐ Histogram
- ☐ Control Chart
- ☐ Box plot
- ☐ Stratification
- ☐ Fishbone diagram
- ☐ Stem-and-leaf plot
- ☐ Scatterplot

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● Question 6

✓ 1 pt ↻ 1

Which of the following are types of Nonparametric Tests?

- ☐ Paired T-test
- ☐ T-test
- ☐ One-way ANOVA
- ☐ Chi-square Test
- ☐ Pearson Correlation
- ☐ Mann-Whitney U Test
- ☐ Wilcoxon Rank Sum Test
- ☐ Kruskal-Wallis Test
- ☐ Wilcoxon Signed Rank Test

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● Question 7

✓ 1 pt ↻ 1

What is a control chart? Briefly describe the steps in creating a control chart and how it is interpreted.

\* Note that this problem can't grade itself and so it'll mark incorrect until I grade it.

